STAT 210A: Theoretical Statistics

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Contents

10/29/26 Lecture 19: Least Favorable Sequences and Hypothesis Testing 1
Definition 19.1: Least Favorable Sequences 1
Theorem 19.2 1
Definition 19.3: Simple and Composite Hypotheses 2
Definition 19.4: Level-α Tests 4

10/29/26 Lecture 19: Least Favorable Sequences and Hypothesis Testing

Recall:

$$r^* = \inf_\delta \sup_\theta R(\theta; \delta)$$

may not be attained, but if $\sup_\theta R(\theta; \delta^*) = r^*$, then δ^* is minimax.

Now, recall

$$r_\Lambda = \inf_\delta \int_\Theta R(\theta, \delta) d\Lambda(\theta)$$

and if Λ^* maximizes r_Λ, then Λ^* is least-favorable. Recall that, for all δ, Λ,

$$r_\Lambda \le r_{\Lambda^*} \le r^* \le \sup_\theta R(\theta, \delta)$$

Thus, if we can find a pair Λ, δ_Λ for which

$$r_\Lambda = \sup_\theta R(\theta, \delta_\Lambda)$$

then Λ is least favorable and δ_Λ is minimax.

Thus, δ is minimax if δ is Bayes with constant risk.

Example: Binomial

$X \sim \text{Binom}(n, \theta)$, δ_Λ is minimax for $\Lambda \sim \text{Beta}\left(\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2}\right)$.

Example: Gaussian

What about $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$, and we want to estimate θ with squared error loss?

Idea: Flat prior is good but not proper.

This motivates LF prior sequences.

Definition 19.1: Least Favorable Sequences

A sequence $\Lambda_1, \Lambda_2, \dots$ is *least favorable* if $r_{\Lambda_k} \to \sup_\Lambda r_\Lambda$.

Theorem 19.2

Suppose $\Lambda_1, \Lambda_2, \dots$ is a sequence of priors and δ is an estimator such that

$$\sup_\theta R(\theta, \delta) = \lim_{n \to \infty} r_{\Lambda_n}$$

Then, $\{\Lambda_i\}$ is LF, δ is minimax, and $r^* = \lim_n r_{\Lambda_n}$.

Proof:

$$r_{\Lambda_i} \le \sup_{\Lambda} r_\Lambda \le r^* \le \sup_\theta R(\theta, \delta)$$

and $r_{\Lambda_i} \to \sup_\theta R(\theta, \delta)$.

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Example: $X \sim N_d(\theta, I_d)$, $\delta_0(X) = X$, and MSE $\equiv d$. Now, we see that $\Lambda_k = N_d(0, kI_d)$ converges to the flat prior. Recall that the Bayes estimator is

$$\delta_{\Lambda_k}(X) = (1 - \zeta_k)X, \quad \zeta_k = \frac{1}{1+k}$$

and so

$$\text{MSE}(\theta, \delta_{\Lambda_k}) = \zeta_k^2 \|\theta\|^2 + (1-\zeta_k)^2 d$$

Taking $k \to \infty$, we see

$$r_{\Lambda_k} = \mathbb{E}[\text{MSE}(\theta, \delta_{\Lambda_k})] = \frac{kd}{(1+k)^2} + \frac{k^2 d}{(1+k)^2} \to d$$

so as $\sup_\theta R(\theta, \delta_0) = d = \lim_{k\to\infty} r_{\Lambda_k}$, we see δ_0 is minimax and $(N_d(0, kI_d))_k$ is LF.

Bounding the Minimax Risk

Notice that our theorem gives us a way to bound r^*, as it tells us that for any δ, Λ,

$$\int R(\theta; \delta_\Lambda) d\Lambda(\theta) \le r^* \le \sup_\theta R(\theta; \delta)$$

thus, although minimax esimators are typically hard to find, minimax bounds are often used to characterize how hard a problem is (especially the lower bound).

That is, we will typically propose a practical estimator δ, and find a Λ for which r_Λ is close to $\sup_\theta R(\theta; \delta)$. Then, if we can do that, we can say that δ cannot be improved much in the minimax sense.

One caveat with minimax estimators however is that a problem might be typically easy throughout most of the parameter space, but very difficult in a bizarre case that is rarely every encountered.

Hypothesis Testing

Setup: Data $X \sim P_\theta$, P_θ in model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$. Partition into:

$$\begin{aligned} \text{Null Hyp.} \quad & H_0 : \theta \in \Theta_0 \\ \text{Alternative Hyp.} \quad & H_1 : \theta \in \Theta_1 \end{aligned}$$

Usually: Require hypotheses to be disjoint and exhaustive (so Θ_0, Θ_1 partition Θ). If Θ_1 is unspecified, we can let $\Theta_1 = \Theta \setminus \Theta_0$.

We have two actions:
Reject H_0: Conclude $\theta \in \Theta_1$
Fail to reject H_0: No conclusion

Typically, P_θ has the same support for all θ, so we can never completely rule out $\theta \in \Theta_0$. Thus, we typically never have complete certainty in our actions.

Definition 19.3: Simple and Composite Hypotheses

H_0 or H_1 are called *simple* if $\Theta_0 = \{\theta_0\}$ or $\Theta_1 = \{\theta_1\}$. Otherwise, we call them *composite*.

Simple hypotheses are nice since we just need to consider one distribution; namely P_{θ_0} or P_{θ_1}.

Example: z-test

$X \sim N(\theta, 1)$.

$$H_0 : \theta \leq 0 \quad \text{vs} \quad H_1 : \theta > 0 \quad \text{(comp vs comp)}$$
$$H_0 : \theta = 0 \quad \text{vs} \quad H_1 : \theta \neq 0 \quad \text{(simple vs comp)}$$

Example: Two-Sample

$X_1, \dots, X_n \overset{\text{iid}}{\sim} P$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} Q$.

$$H_0 : P = Q \quad \text{vs} \quad H_1 : P \neq Q$$

To make these decisions, we construct the critical function

$$\phi(X) = \begin{cases} 0 & \text{accept } H_0 \\ \pi \in (0,1) & \text{reject } H_0 \text{ w.p. } \pi \\ 1 & \text{reject } H_0 \end{cases}$$

If a critical function is not randomized, we partition $\mathcal{X}$ into

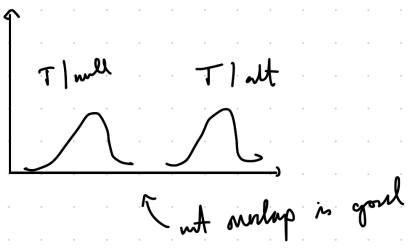
$$\text{rejection region: } R = \{x : \phi(x) = 1\} \qquad \text{acceptance region: } A = \{x : \phi(x) = 0\}$$

A useful way to construct a test is via a *test statistic* $T(X) \in \mathbb{R}$. We say ϕ *rejects for large* $T(X)$ if

$$\phi(x) = \begin{cases} 0 & T(x) < c \\ \gamma \in (0,1) & T(x) = c \\ 1 & T(x) > c \end{cases} \quad \text{(if randomized)}$$

We say c is the *critical threshold*.

Thus, the goal is to choose a good test statistic T to discriminate H_0 from H_1.



Significance Level

Type I Error: H_0 true, but we reject
Type II Error: H_0 false, but we don't reject

Typically we, as statisticians, are required to control Type I, but subject to a constraint on Type I error, we minimize Type II error.

Usual goal: Minimize $\mathbb{P}_{H_1}(\text{Type II})$ subject to $\mathbb{P}_{H_0}(\text{Type I}) \leq \alpha$.

Power function: $\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)] = \mathbb{P}_\theta(\text{Reject } H_0)$. Then, the goal is

$$\underset{\phi}{\text{maximize}}\ \beta_\phi(\theta) \text{ for } \theta \in \Theta_1 \quad \text{subject to} \quad \beta_\phi(\theta) \leq \alpha \text{ for } \theta \in \Theta_0$$

Definition 19.4: Level-α Tests

We say ϕ is a *level-α test* if $\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha$.